Energy conversion

The law of conservation of energy states that energy cannot be destroyed or lost. It can, however, be converted from one form to many other different forms. Electricity is a particularly valued energy source because it can be converted into sound energy, heat (thermal energy), light (radiant energy), and, in the case of a motor, movement (kinetic energy).

Chemical energy
Chemical energy is the energy stored in the bonds of chemical compounds, from foods and batteries to fossil fuels. It can be released through chemical reactions, which break the bonds between atoms. For example, burning a fossil fuel such as coal converts the chemical energy stored in the coal to light and heat.

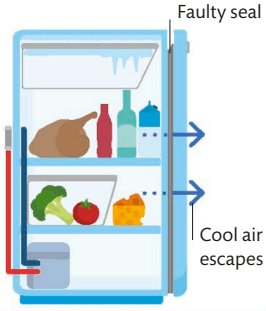
Kinetic energy
Kinetic energy is the energy an object possesses because it is moving, such as a person sprinting or a skier traveling downhill. There are various types of kinetic energy, including rotational and vibrational energy. The amount of kinetic energy an object has depends on its speed and mass.

Mechanical energy
Mechanical energy is the kinetic energy of an object combined with its potential energy—energy that is not doing any work but can be converted—arising due to an object's position. An example is a compressed spring, which releases its potential energy when it bounces back to its original position.

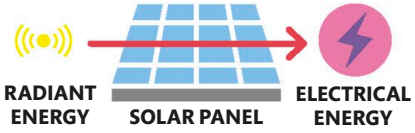
Thermal energy
Thermal energy is technically a type of kinetic energy derived from the vibrating movement of atoms in a substance. Heat describes the flow of thermal energy from one place to another, such as the heat transferred from a flame to a cooking pot on a stove.

WASTED ENERGY

A machine always wastes a proportion of the energy it uses. Light bulbs convert only some of the electricity they receive to light, while some is wasted as heat. A poorly tuned or damaged machine, such as a fridge with a door seal that leaks cool air, can also waste further energy.



Energy conversion in a solar panel
A solar panel contains a series of photovoltaic cells (see p.30). These convert the radiant energy in sunlight into electrical energy in the form of a flow of electrons.





Fossil fuels

Around two-thirds of the world's electricity and more than a billion motor vehicles and other machines are powered by fuel derived from the fossilized remains of once living things. These fossil fuels (oil, coal, and natural gas) are nonrenewable resources with limited reserves. When burned, their chemical energy transforms mostly to heat energy but with significant emissions of greenhouse gases.

CHINA AND THE **US** GENERATE A TOTAL OF **40 PERCENT** OF THE WORLD'S **GREENHOUSE GAS EMISSIONS**